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* Assignment no. 1 *

Q.1)

Aspect	Conventional Energy sources	Non-conventional Energy sources
1) Definition	Traditional energy sources used for many decades	Newer and alternative energy sources.
2) Availability	Limited and exhaustible	Abundant and mostly inexhaustible
3) Pollution	Causes high air and water pollution	Environment-friendly, very low pollution
4) Cost	Lower initial cost but high fuel cost	High initial cost but low running cost
5) Sustainability	Not sustainable for future	Sustainable and suitable for long term use
6) Environmental impact	Severe	Minimal
7) Examples:	Coal, petroleum, natural gas, thermal power	Solar, wind, biomass, tidal, geothermal.

Q.2)	Aspect	Renewable Energy Sources	Non-Renewable Energy Sources
1)	Definition	Energy sources that can be replenished naturally	Energy sources that cannot be replenished
2)	Availability	Unlimited or naturally replenished	Limited and will get exhausted
3)	Environmental impact	Clean and eco-friendly	Causes pollution and global warming.
4)	Operating cost	Low operating and maintenance cost	High fuel and maintenance cost
5)	Future scope	High future potential	Limited future availability
6)	Energy density	Lower	Higher
7)	Long-term use	Suitable for future generations	Not suitable for long-term use.
8)	Examples:	Solar, wind, hydro, biomass	Coal, oil, natural gas, nuclear fuels.

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Q.3)

Aspect	Solar PV System	Solar Thermal System
1) Working principle	Converts sunlight directly into electricity	Converts sunlight into heat energy
2) Main component	Solar photovoltaic cells	Solar collectors
3) Output energy	Electrical energy	Thermal (heat) energy
4) Applications	Lighting, battery charging, grid power	Water heating, space heating, steam generation
5) Efficiency	Lower efficiency (15 - 25 %)	Higher efficiency (30 - 70 %)
6) Energy storage	Stored in batteries	Stored as hot water or thermal fluids.
7) Cost	Higher per unit electricity	Lower for heating applications
8) Maintenance	Low	Moderate.